Visualizing Machine Learning: Illustrated Stable Diffusion

Understanding AI Image Generation with
Stable Diffusion

Introduction to AI Image Generation

Mind-blowing Visuals

- AI image generation technology creates stunning visuals from text descriptions.
- This marks a shift in how humans create art with a magical quality.
- Text encoder of a CLIP model turns input text into vectors for Image Generator.
- Image Generator consists of components that create visuals from text information.

Stable Diffusion Milestone

AI Art Revolution

- StableDiffusion enables high-performance AI image generation for all users.
- Its image information creator component is key to superior performance.
- The model offers top-notch image quality, speed, and low resource requirements.

Versatility of Stable Diffusion

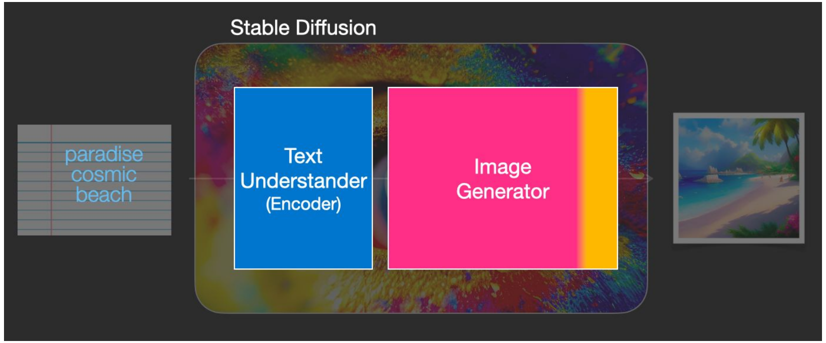
How it works

- Stable Diffusion allows for text-to-image generation and image alteration
- Key components include Stable Diffusion, DallE2, and Google's Imagen
- Images generated without text data, text incorporatio for control described later
- Speed Boost Diffusion improves efficiency and image quality

Components of Stable Diffusion

Under the Hood

- Stable Diffusion is composed of multiple models, not a single entity.
- Text understanding component translates text into numeric representation.
- Diffusion process generates images without using text data.
- Incorporating text allows control over the type of image generated.



Text Encoder in Action

Transformer Language Model

- The text encoder is a key component of the CLIP model, providing vector representations for words or tokens.
- These vectors are then utilized by the Image Generator to create images based on text prompts.
- The conditioning components in Figure 3 of the LDMStable Diffusion paper show how text guides image generation.
- The Transformer language model enables language understanding in the process of generating images.

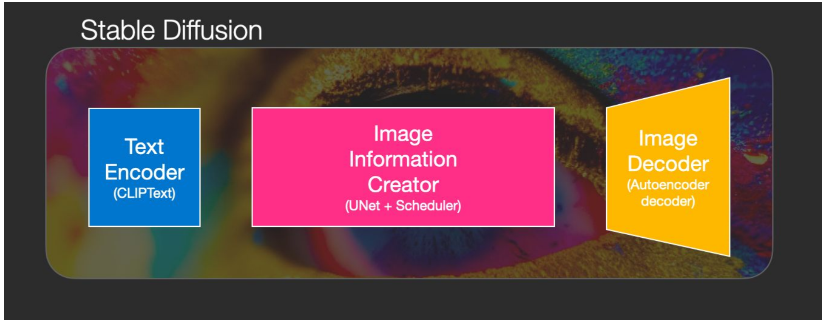
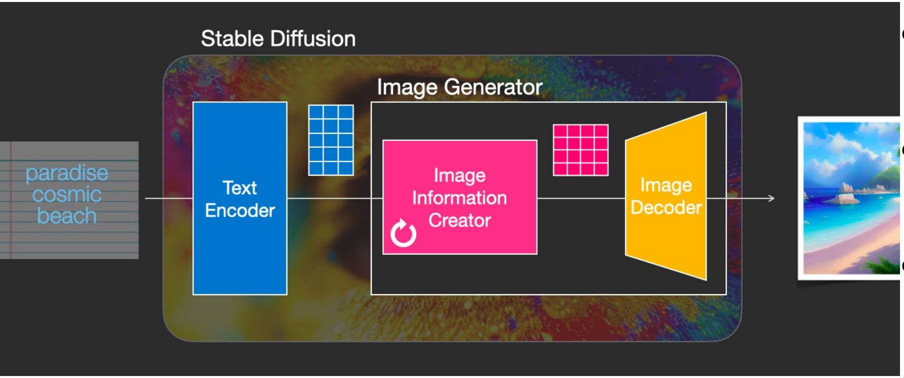


Image Generator Overview



Two Stage Process

- The image information creator is the key component in Stable Diffusion for performance gains.
- Multiple steps are run to generate image information, usually defaulting to 50 or 100 steps.
- Diffusion models frame image generation as a problem solvable by powerful computer vision models.
- Complex operations can be learned with large datasets, resulting in detailed images.

Image Information Creator

Efficient Image Generation

- The image information creator is the key component in Stable Diffusion for performance gain.
- It runs for multiple steps to generate image information, often defaulting to 50 or 100 steps.
- The image decoder then paints the final pixel image from the information created by the creator.
- Stable Diffusion consists of three main components: ClipText, UNet, and Scheduler.

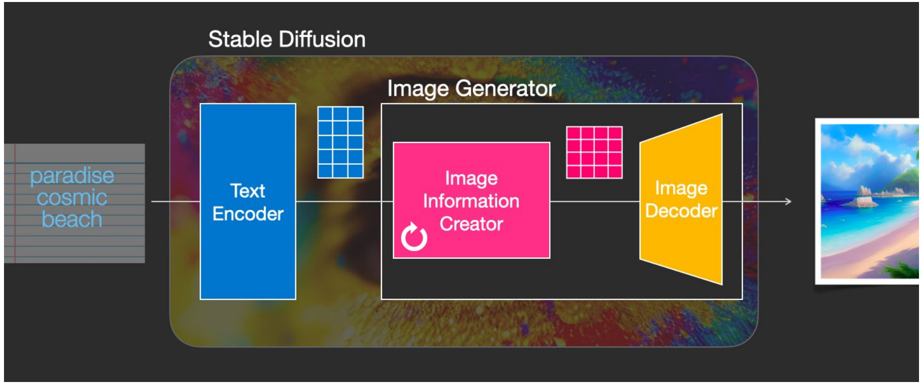
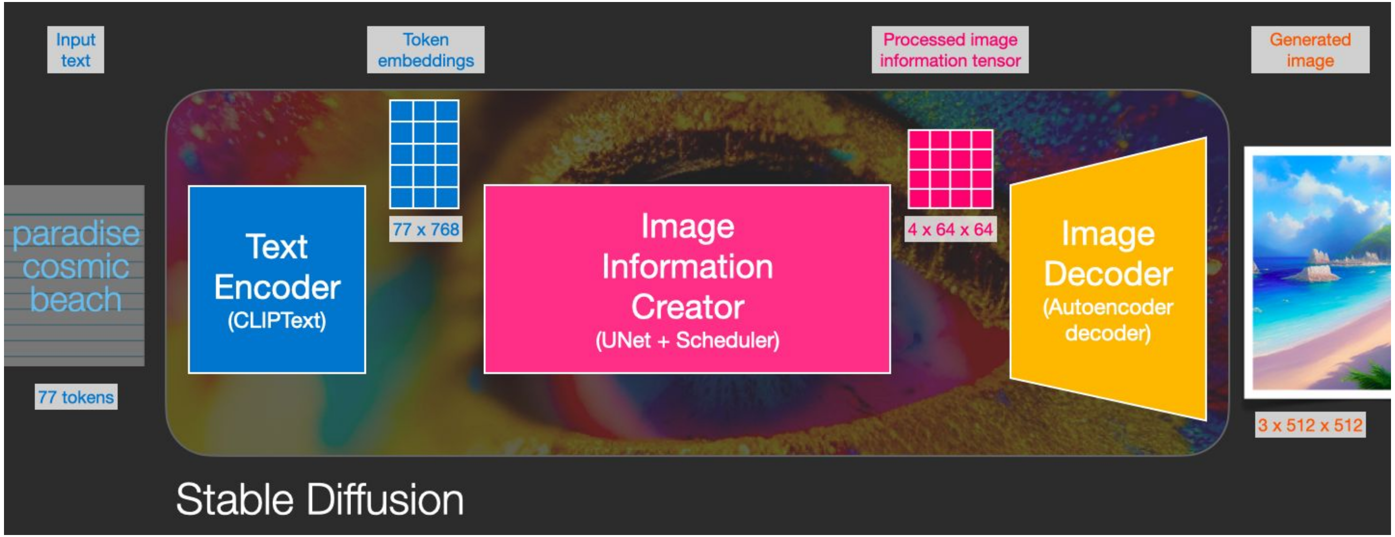


Image Generation Process

Step 2-4 Insight

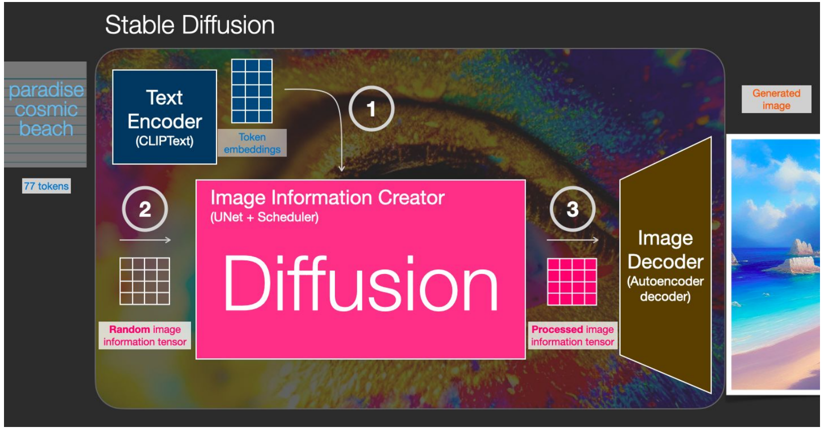
- Between steps 2 and 4, the outline emerges from noise, creating a fascinating transformation.
- Diffusion models utilize powerful computer vision models to learn complex operations efficiently.



What is Diffusion Anyway?

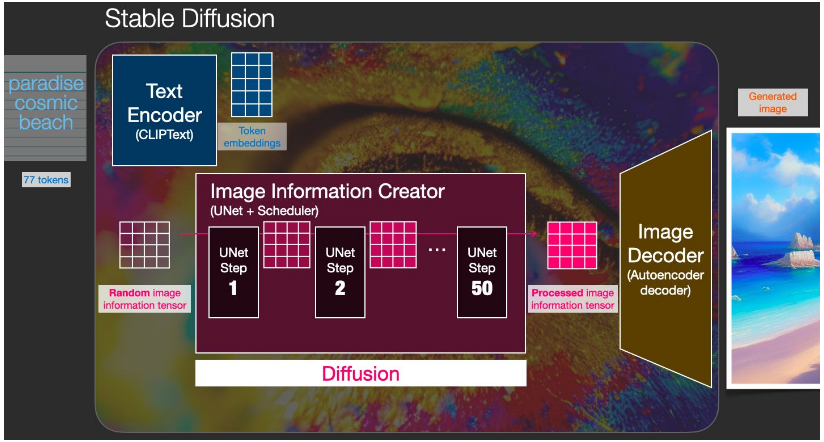
Diffusion Models

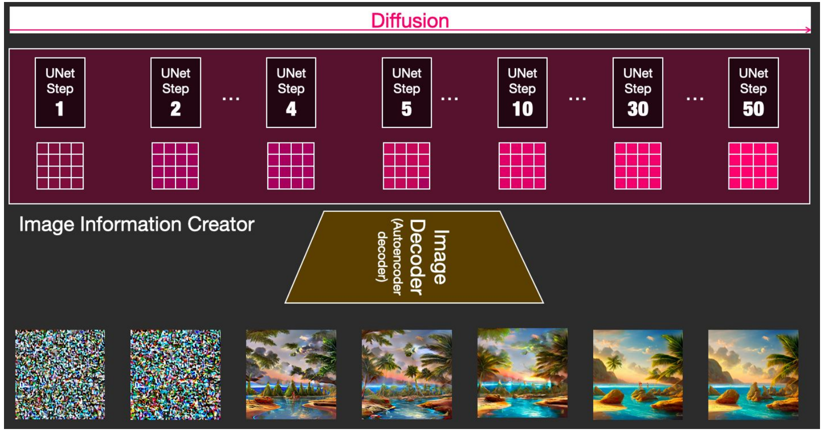
Diffusion is the process that takes place inside the "image information creator" component. Having the token embeddings that represent the input text, and a random starting *image information array* (these are also called *latents*), the process produces an information array that the image decoder uses to paint the final image.



Diffusion Steps

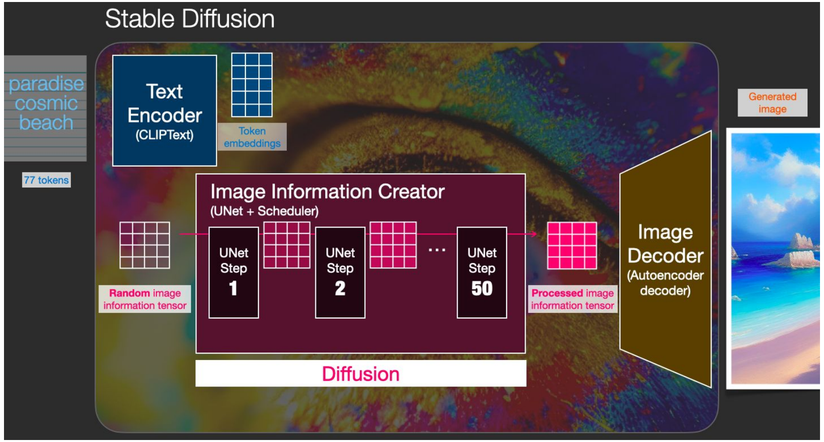
Each step operates on an input latents array, and produces another latents array that better resembles the input text and all the visual information the model picked up from all images the model was trained on.

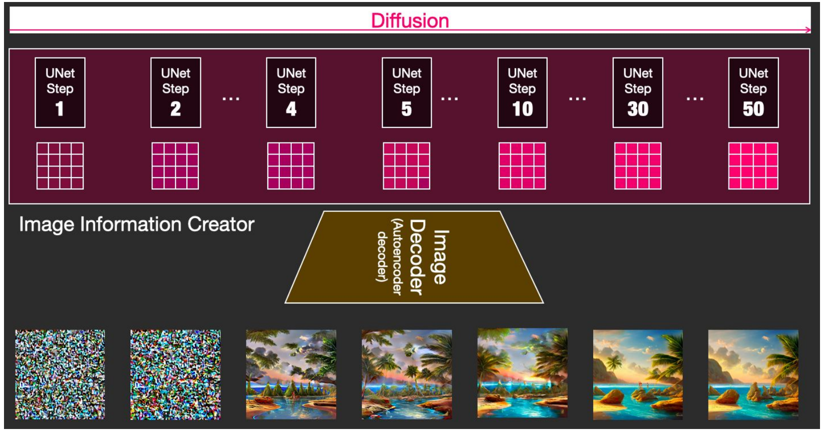




Diffusion Steps

Each step operates on an input latents array, and produces another latents array that better resembles the input text and all the visual information the model picked up from all images the model was trained on.





Resources

- A one-minute YouTube short on using Dream Studio to generate images with Stable Diffusion.
- How does Stable Diffusion work? – Latent Diffusion Models EXPLAINED [Video]
- Stable Diffusion - What, Why, How? [Video]